

Lab 04 - SOP/POS and KMaps

In this lab, you've learned how to apply KMaps, Sum Of Products and Products of sums to simplify digital logic equations. Then, you've proven out that they work using an implemented design on your Basys3 boards.

Rubric

Item	Description	Value
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Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

In this lab we learned how to use a Kmap to find both the minterms and maxterms of a circuit using sum of product, and product of sums. We then wrote it into a verilog file, and compared the minterms and maxterms to the original table to prove they were equivalent. This shows how much space can be saved by simplifying down to the minterms, and the Kmap made it simple to find the correct terms.

Lab Questions

Why are the groups of 1's (or 0's) that we select in the KMap able to go across edges? The groups are able to go across edges because only 1 variable is changing when you move across the grid. This is possible because we swap the 11 and 10 rows/columns which allow us to visualize the matrix like a cylinder.

Why are the names Sum of Products and Products of Sums?

AND is a product of two terms, and OR is a sum of two terms.

The sum of products is taking ANDs and linking them together with OR's. This shows the name Sum of products since it is the sum of multiple products linked together.

(A & B) | (C & D)

The product of sums is multiple ORs linked together with ANDs. This is the product of multiple sums and gives the name. (A | B) & (C | D)

Open the test.v file – how are we able to check that the signals match using XOR? The test.v doesn't know whether the output for any given row is supposed to be 1 or 0. By using exclusive or it checks for a mismatch of outputs, fails if they do not match, and passes as long as both outputs are the same. 0 & 0 pass, 1 & 1 pass, 0 & 1 fail, 1 & 0 fail.